

Alexandru COZMA
Data Engineer | Big Data & ETL

in [linkedin.com/in/Alexandru_Cozma](https://www.linkedin.com/in/Alexandru_Cozma) +49 157 81770036 @ alex.czma@proton.me
📍 Amalienstraße 77, 76133 Karlsruhe, Germany

Data Engineer experienced in building robust, distributed data pipelines and integrating heterogeneous data sources. Strong in Python (PySpark), SQL and data modeling, comfortable with CI/CD and automated testing, and a fast learner who quickly adapts to new, complex data ecosystems.



🎓 EDUCATION

04.2026	B.Sc. Data Science, KARLSRUHE UNIVERSITY OF APPLIED SCIENCES,
09.2022	Focus areas : Database Systems, Data Engineering, Machine Learning, Data Science, Business Intelligence
	➤ Time Series Analysis (2023) Built distributed, automated data & ML pipelines with PySpark to forecast restaurant revenue; designed and persisted the results in a cloud data warehouse (Snowflake) on Google Cloud Platform
	PySpark Snowflake Google Cloud Platform Distributed Systems Forecasting
	➤ Client Project - Causal Analysis & Clustering (2023 - 2024) Causal analysis and development of a clustering pipeline for text and image embeddings, in close collaboration with the client; the derived measures contributed to a revenue increase of around 10 % for the client 🔗 Repository
	Python NLP Computer Vision Embeddings Scrum Kanban
	➤ Machine Learning - Tree Species Classification (2024) Processing and classification of satellite time-series data (multidimensional geospatial data) to determine tree species using machine learning methods 🔗 Repository
	Python scikit-learn Data Exploration Satellite Data Geodata

🏢 EXPERIENCE

04.2026	Bachelor Thesis – Data Engineering & Data Integration, CONSILIARI GMBH, Karlsruhe
11.2025	➤ Designed and built an end-to-end data pipeline (ingestion, transformation, indexing) to process large volumes of heterogeneous, unstructured data with PostgreSQL (pgvector)
	➤ Ensured data quality and lineage through automated unit tests and validation concepts for downstream AI models
	➤ Designed efficient chunking and query strategies for performant, resource-efficient processing of large datasets
	Data Pipelines Data Quality Data Modeling Unstructured Data PostgreSQL Python Docker Git CI/CD
02.2025	Data Warehouse Engineer Intern, OESTERREICHISCHE KONTROLLBANK AG, Vienna
09.2024	➤ Automated daily ETL processes for large data volumes in a regulated environment, working closely with business departments to translate requirements into production-ready data structures
	➤ Drastically reduced manual data processing while ensuring data quality and traceability for financial reporting
	➤ Designed robust data models for the data warehouse and developed and optimized complex SQL processes
	ETL Data Warehousing Data Modeling SQL Oracle DB Cross-functional
07.2024	Working Student Data Analyst / Cyber Security, SIEMENS AG, Karlsruhe
08.2023	➤ Built an interactive reporting and visualization solution (Python, Power BI) at the interface between IT and business, from requirements gathering to production
	➤ Independently prepared and presented analytical results to stakeholders in an agile project environment
	Python Power BI Data Analysis Stakeholder Communication Agile

☰ SKILLS

Data Engineering	ETL/ELT Pipelines, Spark (PySpark), Data Modeling, Data Warehousing, Data Quality & Validation
Programming & Tools	Python 3 (Pandas, NumPy, PySpark), SQL (advanced), C#, Git/GitHub, Docker, CI/CD, pytest
Cloud & Databases	Google Cloud Platform, Snowflake, PostgreSQL (pgvector), Oracle DB, Azure & AWS (basics)
Data Science & ML	scikit-learn, PyTorch, RAG & LLMs

🗣️ LANGUAGES

German ●●●●●
English ●●●●○
Spanish ●●●●○

+ OTHER

➤ Driving licence (Class B)